Neural Network-Aided Kalman Filtering for Indoor Positioning Autonomous Vehicle System

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***Abstract*—Indoor positioning systems (IPS) are essential for autonomous vehicles in GPS-denied environments. While Ultra-Wideband (UWB) ranging offers high precision, multipath propagation and non-line-of-sight (NLOS) interference cause irregular scatter that degrades trajectory stability. Conventional Kalman filtering only partially suppresses this noise due to its reliance on fixed, hand-tuned covariance matrices that cannot adapt to spatially varying errors. This paper presents a neural network-aided Kalman filtering framework for UWB localization using DWM1001C hardware. The system converts UWB ranges into 2D positions via Least Squares Estimation, then employs a recurrent neural network to dynamically compute a data-driven Kalman gain. This implicitly learns time-varying noise characteristics while preserving the real-time recursive structure of the standard filter. Evaluated over 60,971 samples across eight dynamic trajectories, the proposed method retains 79.4% of position estimates within ± 200 mm of the reference path, outperforming raw Least Squares (55.1%) and the conventional Kalman filter (56.5%). At wider tolerances, it achieves 99.0% within ± 500 mm and 100.0% within ± 1000 mm. The overall RMSE is reduced to 146.5 mm—a $\sim$47% improvement over both baselines. These results demonstrate significant improvements in accuracy, trajectory smoothness, and robustness under practical indoor UWB conditions.**

I. Introduction

Accurate indoor positioning systems (IPS) are increasingly essential for autonomous vehicles, mobile robots, smart buildings, retail services, and augmented-reality applications [1]. Because global navigation satellite system (GNSS) signals are severely distorted indoors, they cannot provide the reliability required for autonomous operations, motivating the development of robust IPS technologies.

Among explored IPS technologies—including Wi-Fi [2], Bluetooth Low Energy (BLE) [3], [17], and Ultra-Wideband (UWB) [4], [5]—UWB is uniquely attractive. Its wider bandwidth provides centimeter-level ranging and strong robustness against narrowband interference. Compared to Wi-Fi and BLE, UWB is better suited for autonomous-vehicle positioning where accuracy and stability are critical [6]. However, practical UWB localization remains challenged by multipath propagation, non-line-of-sight (NLOS) conditions, antenna-delay bias, anchor-geometry errors, clock drift, and fast motion [7].

Prior studies address these limitations from various perspectives. Jiang *et al.* [8] integrated UWB, GNSS, and inertial navigation for seamless indoor–outdoor positioning. Wang *et al.* [9] proposed NLOS correction via LOS/NLOS mapping, and Qi *et al.* [10] studied anchor position calibration. Sidorenko *et al.* [11] analyzed ranging-error correction. Collectively, these works highlight that UWB accuracy depends heavily on environmental awareness and measurement post-processing.

Filtering offers a complementary approach to improve stability. Kalman Filter (KF) methods provide computationally efficient, recursive state estimation for embedded systems [12], [13], while the Extended Kalman Filter (EKF) linearizes nonlinear observations. Feng *et al.* [14] showed integrating UWB and inertial data via KF improves accuracy, and Liu *et al.* [15] demonstrated feasible-region filtering enhances indoor stability. However, conventional KF and EKF performance relies on predefined motion models, static noise covariances, and linearization assumptions, which often fail to capture complex autonomous vehicle dynamics in cluttered environments.

To address these challenges, a neural network-aided Kalman filtering framework is proposed for UWB localization in autonomous vehicles. The main contributions of this paper are summarized as follows:

1) A novel neural network-aided Kalman filter is developed for UWB-based localization. This framework preserves the recursive interpretability of classical filters while learning a dynamic, data-driven gain to improve positioning accuracy, trajectory smoothness, and robustness in practical indoor de-ployments.
2) We evaluate raw LSE, a tuned conventional KF, and the proposed NN-aided KF using the same eight rectangular-loop recordings comprising 60,971 samples. Performance is assessed through path-referenced RMSE and the percentage of estimates within predefined acceptable error range

The remainder of this paper is organized as follows. Section II details the localization methods, including LSE, the conventional KF, and the proposed Neural Network-Aided Kalman Filter. Section III describes the localization system and hardware configuration. Section IV reports the experimental setup and evaluation results. Section V concludes the paper and outlines future work.

II. Localization Methods

A. Least Squares Estimation

Because the anchors are installed at a height $h = 1.4$ m above the ground, the measured tag-to-anchor distance $d_{i,tag}$ must be converted to the two-dimensional ground-plane distance d_i by the Pythagorean relation

$$d_i = \sqrt{d_{i,tag}^2 - h^2}. \tag{1}$$

The Least Squares Estimation (LSE) algorithm is then used to estimate the initial two-dimensional position from the four corrected distances d_i. This initial position is used as a reliable input for both Kalman filter and the proposed Neural Network aided Kalman Filter

$$\mathbf{A}\mathbf{x}_{LS} = \mathbf{b}, \tag{2}$$

where

$$\mathbf{A} = \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \\ \vdots & \vdots \\ x_N - x_1 & y_N - y_1 \end{bmatrix}, \quad \mathbf{x}_{LS} = \begin{bmatrix} x_{LS} \\ y_{LS} \end{bmatrix},$$
$$\mathbf{b} = \frac{1}{2} \begin{bmatrix} x_2^2 + y_2^2 - d_2^2 - (x_1^2 + y_1^2 - d_1^2) \\ x_3^2 + y_3^2 - d_3^2 - (x_1^2 + y_1^2 - d_1^2) \\ \vdots \\ x_N^2 + y_N^2 - d_N^2 - (x_1^2 + y_1^2 - d_1^2) \end{bmatrix}. \tag{3}$$

The estimated coordinate is obtained by solving

$$\mathbf{x}_{LS} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}. \tag{4}$$

B. Kalman Filter Algorithm

Consider a system with a position-velocity model in two-dimensional where the velocities along the axis (x, y) are denoted by $(\dot{x}, \dot{y})$. The system state at time step k can be written as

$$\mathbf{x}_k = [x_k,\, y_k,\, \dot{x}_k,\, \dot{y}_k]^\top, \tag{5}$$

where (x_k, y_k) is the planar robot position and $(\dot{x}_k, \dot{y}_k)$ represents velocity along the x- and y-axes. The previous posterior estimate is propagated to the current time step using

$$\hat{\mathbf{x}}_{k|k-1} = F\hat{\mathbf{x}}_{k-1|k-1}, \tag{6}$$

where the constant-velocity transition matrix is

$$F = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{7}$$

The corresponding prior covariance matrix is computed as

$$P_{k|k-1} = FP_{k-1|k-1}F^\top + Q, \tag{8}$$

where Q is the process-noise covariance matrix.

In the correction step, the predicted state is updated using the pre-processed 2D position estimate $\mathbf{z}_k = [x_{lse,k},\, y_{lse,k}]^\top$ derived from the UWB Least Squares Estimation. Since the measurement is already expressed as a Cartesian position, the observation model is linear and defined as

$$h(\hat{\mathbf{x}}_{k|k-1}) = H\hat{\mathbf{x}}_{k|k-1}, \tag{9}$$

where the constant observation matrix H extracts only the position components from the state vector:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \tag{10}$$

The recursive prediction–correction procedure smooths the UWB-derived position estimates while preserving real-time operation of the system. Algorithm 1 summarizes the complete recursive implementation, including initialization, prediction, gain computation, and correction steps.

C. Neural Network-Aided Kalman Filter

The proposed stabilization algorithm replicates the idea of KalmanNet [16]: the recursive structure of the Kalman filter is preserved, while a neural network learns the Kalman gain from data. This is suitable for the proposed UWB vehicle system because the state-transition and observation models are partially known, but the true noise statistics and NLOS-induced errors are difficult to model analytically. Fig. 1 presents the overall neural network-aided Kalman filtering architecture, while Fig. 2 details the recurrent neural network used to estimate the adaptive gain.

Algorithm 1: Kalman Filter

Input: Pre-processed 2D position measurements $\mathbf{z}_k = (x_{lse}, y_{lse})$;
Starting position $\mathbf{z}_0 = (x_0, y_0)$;
Sampling interval $\Delta t = 0.0125$ s (80 Hz)

Initialize: State vector $\hat{\mathbf{x}}_{0|0} = [x_0,\, y_0,\, 0,\, 0]^\top$ with zero initial velocity;
State transition matrix $\mathbf{F}$ for the constant-velocity model with Δt;
Observation matrix $\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$;
Initial state covariance $\mathbf{P}_{0|0}$ is set to $\mathrm{diag}(500^2,\, 500^2,\, 200^2,\, 200^2)$;
Tuned Parameters: $\mathbf{Q}$ and $\mathbf{R}$ are selected by grid search to minimize cross-track RMSE;
$\mathbf{Q} = \mathrm{diag}(100,\, 100,\, 10,\, 10)$ and
$\mathbf{R} = \mathrm{diag}(100^2,\, 100^2)$.

while *new measurement* $\mathbf{z}_k$ *is available* **do**
  Predict the state: $\hat{\mathbf{x}}_{k|k-1} \leftarrow \mathbf{F}\hat{\mathbf{x}}_{k-1|k-1}$;
  Predict the covariance: $\mathbf{P}_{k|k-1} \leftarrow \mathbf{F}\mathbf{P}_{k-1|k-1}\mathbf{F}^\top + \mathbf{Q}$;
  Compute the innovation: $\mathbf{y}_k \leftarrow \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}$;
  Compute the Kalman gain: $\mathbf{K}_k \leftarrow \mathbf{P}_{k|k-1}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\top + \mathbf{R})^{-1}$;
  Update the state: $\hat{\mathbf{x}}_{k|k} \leftarrow \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k\mathbf{y}_k$;
  Update the covariance: $\mathbf{P}_{k|k} \leftarrow (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1}$;
end

Output : Smoothed position estimate $[\hat{x}_{k|k}, \hat{y}_{k|k}]$

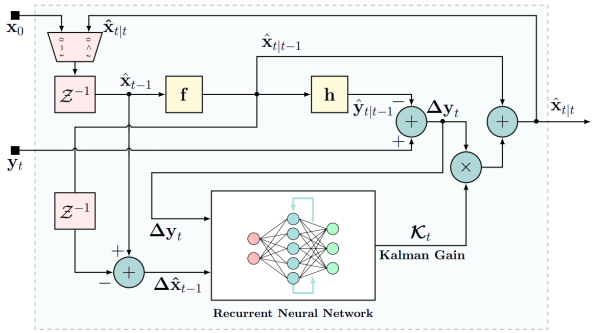


Fig. 1. Neural network-aided Kalman filtering architecture.

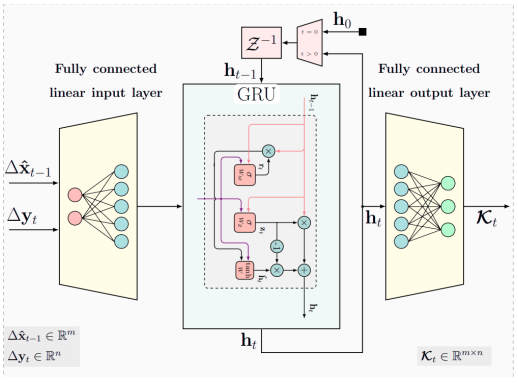


Fig. 2. Recurrent Neural Network architecture.

1) Network Architecture and Feature Extraction: Crucially, while the conventional Kalman filter operates on the UWB-derived position $\mathbf{z}_k = [x_{lse,k},\, y_{lse,k}]^\top$ using static covariance matrices, the neural network learns to dynamically adjust the Kalman gain by extracting temporal kinematic context directly from the UWB sequence. Without relying on external IMU or odometry data, the network analyzes the history of positional updates. If the UWB position jumps drastically but the temporal history suggests the robot should be moving smoothly, the neural network learns to recognize the UWB spike as multipath noise and reduces the Kalman gain accordingly to reject the outlier.

The network $g_\theta(\cdot)$ is designed to extract these relevant physical dynamics and filter out observation noise using a three-stage architecture. Before entering the network, an 8-dimensional feature vector $\mathbf{e}_k$ is constructed by concatenating four distinct differences (each normalized to prevent gradient explosion):

1) **Observation Difference** ($\Delta\mathbf{z}_k \in \mathbb{R}^2$): The step-wise change in the raw UWB measurements ($\mathbf{z}_k - \mathbf{z}_{k-1}$), capturing the instantaneous measured velocity.
2) **Innovation** ($\mathbf{y}_k \in \mathbb{R}^2$): The residual between the current measurement and the model's prediction. A sudden spike here alerts the network to potential UWB multipath errors.
3) **Evolution Difference** ($\Delta\hat{\mathbf{x}}_k^{\text{post}} \in \mathbb{R}^2$): The change in the posterior state estimate ($\hat{\mathbf{x}}_{k|k} - \hat{\mathbf{x}}_{k-1|k-1}$), capturing the smoothed trajectory trend.
4) **Update Difference** ($\Delta\hat{\mathbf{x}}_k^{\text{upd}} \in \mathbb{R}^2$): The correction magnitude applied by the previous update step ($\hat{\mathbf{x}}_{k|k} - \hat{\mathbf{x}}_{k|k-1}$).

First, a fully connected input layer projects this 8-dimensional feature vector $\mathbf{e}_k$ to a 640-dimensional latent representation, followed by a Rectified Linear Unit (ReLU) activation:

$$\mathbf{h}_k^{(0)} = \mathrm{ReLU}(W_{in}\,\mathbf{e}_k + \mathbf{b}_{in}), \quad W_{in} \in \mathbb{R}^{640\times 8}. \tag{11}$$

Second, two stacked Gated Recurrent Unit (GRU) layers with a hidden dimension of 640 process the temporal sequence. Unlike memoryless feed-forward layers, the GRU's persistent hidden state $\mathbf{c}_{k-1}$ carries historical information forward. This allows the network to capture kinematic dependencies across time steps and distinguish between genuine robot movement and physically inconsistent UWB scatter:

$$\mathbf{h}_k^{(l)},\, \mathbf{c}_k^{(l)} = \mathrm{GRU}^{(l)}\Big(\mathbf{h}_k^{(l-1)},\, \mathbf{c}_{k-1}^{(l)}\Big), \quad l \in \{1,2\}. \tag{12}$$

Third, a linear output layer maps the final hidden state to an eight-dimensional vector, which is reshaped into the 4×2 Kalman gain matrix:

$$\begin{aligned} &K_k^{\text{NN}} = \mathrm{reshape}\Big(W_{out}\,\mathbf{h}_k^{(2)} + \mathbf{b}_{out}\Big),\\ &K_k^{\text{NN}} \in \mathbb{R}^{4\times 2},\\ &W_{out} \in \mathbb{R}^{8\times 640}. \end{aligned} \tag{13}$$

The total number of trainable parameters across all layers is 4,933,768.

2) Training Procedure: The network is trained in a supervised manner using trajectories collected from the vehicle. The dataset is partitioned as follows: trajectories from three independent collection sessions are used for training, a fourth independently collected session serves as the validation set for hyperparameter tuning and early stopping, and the final unseen evaluation set consists of eight rectangular-loop trajectories in the target deployment environment.

Since the robot operates on a predefined track with known geometry, the reference state $\mathbf{x}_k^{\text{ref}} = [x_k^{\text{ref}},\, y_k^{\text{ref}}]^\top$ at each time step k is obtained by projecting the raw UWB-derived position onto the nearest point along the ideal ground truth path, defined by the set of known track waypoints. This calculated ground truth serves as the supervision signal in place of an external localization reference.

The Adam optimizer with L_2 weight decay regularization is employed to minimize the mean-squared state-estimation error over a trajectory of length T:

$$\mathcal{L}(\theta) = \frac{1}{T}\sum_{k=1}^{T} \big\|\mathbf{x}_k^{\text{ref}} - \hat{\mathbf{x}}_{k|k}(\theta)\big\|_2^2, \tag{14}$$

where $\mathbf{x}_k^{\text{ref}}$ denotes the ground truth reference state at time step k.

After training, only the forward pass of $g_\theta(\cdot)$ is required online. Thus, the method retains the interpretability and recursive operation of Kalman filtering while allowing the gain K_k^{NN} to adapt implicitly to UWB measurement errors, model mismatch, and indoor multipath effects — without requiring explicit knowledge of Q or R. Algorithm 2 summarizes the online neural network-aided filtering procedure.

Algorithm 2: Neural Network-Aided Kalman Filter

Input: Pre-processed 2D position measurements $\mathbf{z}_k = [x_{lse,k},\, y_{lse,k}]^\top$;
Starting position $\mathbf{z}_0 = (x_0, y_0)$;
Sampling interval $\Delta t = 0.02$ s (80 Hz);
Pre-trained network parameters θ

Initialize: State vector $\hat{\mathbf{x}}_{0|0} = [x_0,\, y_0,\, 0,\, 0]^\top$ with zero initial velocity;
State transition matrix $\mathbf{F}$ for the constant-velocity model with Δt;
Observation matrix $\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$;
GRU hidden states $\mathbf{c}_0^{(1)} \leftarrow \mathbf{0}$ and $\mathbf{c}_0^{(2)} \leftarrow \mathbf{0}$.

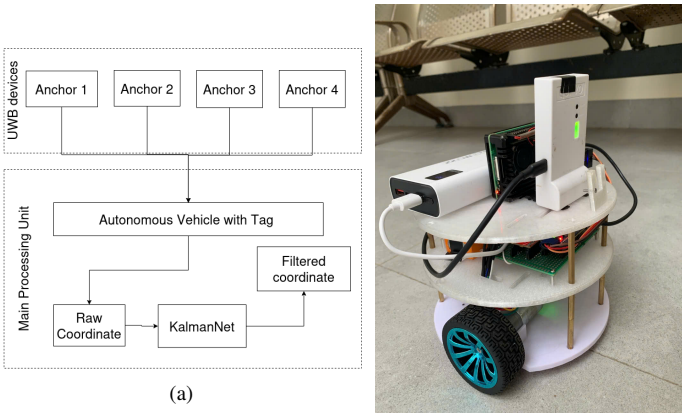
while *new measurement* $\mathbf{z}_k$ *is available* **do**
  Predict the state: $\hat{\mathbf{x}}_{k|k-1} \leftarrow \mathbf{F}\hat{\mathbf{x}}_{k-1|k-1}$;
  Compute the innovation: $\mathbf{y}_k \leftarrow \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}$;
  Construct 8D feature vector $\mathbf{e}_k$ from $\mathbf{y}_k$, $\Delta\mathbf{z}_k$, $\Delta\hat{\mathbf{x}}_{k-1}^{\text{post}}$, and $\Delta\hat{\mathbf{x}}_{k-1}^{\text{upd}}$;
  Input layer: $\mathbf{h}_k^{(0)} \leftarrow \mathrm{ReLU}(W_{in}\mathbf{e}_k + \mathbf{b}_{in})$;
  First GRU layer:
  $\mathbf{h}_k^{(1)}, \mathbf{c}_k^{(1)} \leftarrow \mathrm{GRU}^{(1)}(\mathbf{h}_k^{(0)}, \mathbf{c}_{k-1}^{(1)})$;
  Second GRU layer:
  $\mathbf{h}_k^{(2)}, \mathbf{c}_k^{(2)} \leftarrow \mathrm{GRU}^{(2)}(\mathbf{h}_k^{(1)}, \mathbf{c}_{k-1}^{(2)})$;
  Estimate data-driven gain:
  $\mathbf{K}_k^{\text{NN}} \leftarrow \mathrm{reshape}(W_{out}\mathbf{h}_k^{(2)} + \mathbf{b}_{out})$;
  Update the state: $\hat{\mathbf{x}}_{k|k} \leftarrow \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k^{\text{NN}}\mathbf{y}_k$;
end

Output : Smoothed position estimate $[\hat{x}_{k|k}, \hat{y}_{k|k}]$

III. Localization System

The positioning subsystem estimates the vehicle position from UWB range measurements obtained between one mo-

bile DWM1001C tag mounted on the robot and four fixed DWM1001C anchors installed at known coordinates. Four modules are configured as fixed reference nodes (Anchors) at known locations, while one operates as a mobile target (Tag) equipped on the AMR. The anchors and tag perform continuous Two- Way Ranging (TWR) by exchanging timestamped packets to estimate distances from time-of-flight information. By exchanging UWB packets, the tag obtains time-of-arrival information and converts it into distance measurements. During each ranging cycle, four tag-to-anchor distances are collected, corrected using the known anchor–tag height difference, and forwarded to the localization and filtering modules. The complete hardware and data-flow structure is shown in Fig. 3.



(a) (b)

Fig. 3. System architecture: (a) UWB system model; (b) Our AMR.

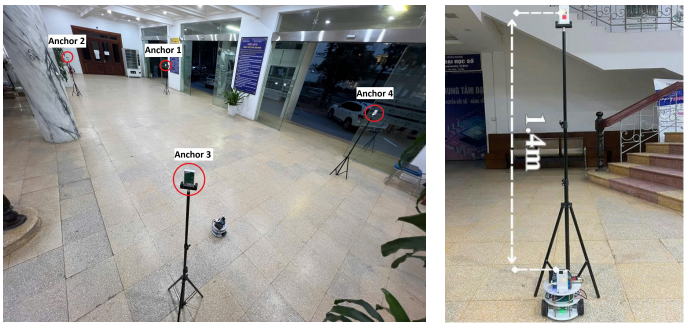
The system is divided into a sensor processing block and a localization processing block. The experimental AMR platform used for data collection is shown in Fig. 3. The sensor processing block includes an STM32F103C8T6 microcontroller, a BNO055 9-axis inertial sensor, and a magnetic encoder for reading onboard motion information and controlling the vehicle. The localization processing block includes a Raspberry Pi 4 and a DWM1001C UWB module. The Raspberry Pi 4 receives UWB distance data, runs the proposed neural network-aided Kalman filtering algorithm, and provides the estimated vehicle state to the navigation modules.

As represented in Fig. 3, the Raspberry Pi 4 functions as the central processing unit. It acquires distance data from the DWM1001C tag and synchronized motion information from the sensor processing block to estimate the vehicle position. Based on the estimated state, the Raspberry Pi 4 sends control commands to the STM32F103C8T6 microcontroller to regulate vehicle movement. This procedure runs continuously through UART communication, enabling the robot to update localization information and maintain stable operation in real time.

IV. Experimental Evaluation

A. Environment and Data Acquisition

The experiment was conducted in an indoor LOS environment at G2 building, VNU University of Engineering and Technology. Four DWM1001C UWB anchors were fixed at ground-plane coordinates $A3 = (0, 0)$, $A4 = (4000, 0)$, $A1 = (4000, 8800)$, and $A2 = (0, 8800)$ mm with a height separation of 1400 mm above the mobile tag. The robot carried the UWB tag, a magnetic encoder, and a BNO055 IMU. All measurements were synchronized and recorded at 80 Hz. Fig. 4 shows the indoor LOS environment and the physical anchor installation used in the experiments.



(a) (b)

Fig. 4. Experimental setup: (a) indoor LOS environment; (b) anchor installation.

The evaluation dataset consists of eight independent recording sessions (Data 1–8), each a complete rectangular-loop traversal along $A3 \rightarrow A4 \rightarrow A1 \rightarrow A2 \rightarrow A1$ with a mean robot speed of approximately 0.18 m/s. The KalmanNet model was trained and validated on recordings from four earlier independent sessions and was never exposed to Data 1–8 during training, ensuring a strict temporal train/test separation. The sampling rate for each file are 80Hz with a total of 60,971 samples.

Since no external localization infrastructure (e.g., motion capture or LiDAR-SLAM) is available, the ground-truth reference path $\mathcal{P}$ is defined analytically as the piecewise-linear polygon connecting the four known waypoints in sequence:

$$\mathcal{P} = \overline{AB} \cup \overline{BC} \cup \overline{CD} \cup \overline{DA}. \tag{15}$$

This geometric path is fully determined by the physical layout of the track and serves as the reference against which all filter outputs are evaluated. At each time step k, the reference position $\mathbf{x}_k^{\text{ref}}$ is defined as the closest point on $\mathcal{P}$ to the raw UWB-derived measurement:

$$\mathbf{x}_k^{\text{ref}} = \underset{\mathbf{p} \in \mathcal{P}}{\arg\min} \ \|\hat{\mathbf{z}}_k - \mathbf{p}\|_2 . \tag{16}$$

quantitative evaluation. This calculated reference is used as the supervision signal during KalmanNet training

Three positioning outputs are generated from the same synchronized recording and compared against $\mathcal{P}$: the raw Least Squares (LSE) trajectory, the conventional Kalman-filtered trajectory, and the proposed KalmanNet trajectory. Using identical timestamps, anchor geometry, height correction, and initialization samples ensures that any performance difference is attributable solely to the filtering strategy. Fig. 5 shows the rectangular reference path and UWB anchor configuration used for this comparison.

The main evaluation method is based on the acceptable range of positioning error. This criterion is well suited to the autonomous vehicle scenario because the vehicle does not only require a low average error; it must keep most estimated positions within a safe ± 200 mm acceptable error range around the planned path. The position error at each time step is computed as the Euclidean distance between the estimated position and the corresponding reference point on the rectangular trajectory,

$$e_k = \sqrt{(x_k - x_k^{\text{ref}})^2 + (y_k - y_k^{\text{ref}})^2}. \tag{17}$$

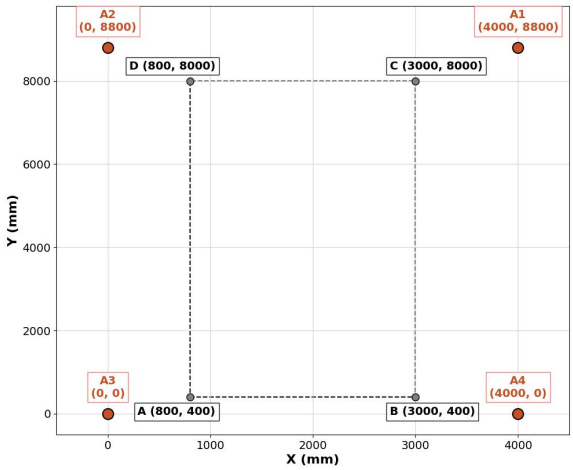


Fig. 5. Rectangular reference trajectory and UWB anchor configuration.

An acceptable error range is then defined around the reference trajectory, and each estimated point is classified as acceptable when

$$e_k \leq e_{\text{acc}}, \tag{18}$$

where $e_{\text{acc}} = 200$ mm is the maximum allowable positioning deviation for the indoor vehicle task. The primary performance score is the acceptable tracking ratio,

$$\eta_{\text{acc}} = \frac{1}{K} \sum_{k=1}^{K} \mathbb{I}(e_k \leq e_{\text{acc}}) \times 100\%, \tag{19}$$

where K is the number of synchronized samples and $\mathbb{I}(\cdot)$ is an indicator function. A higher η_{acc} indicates that more localization samples remain inside the acceptable error range and that the estimated path is more reliable for vehicle navigation.

The trajectory plot is used as a qualitative metric to verify whether each method follows the expected rectangular route and whether oscillatory or scattered estimates are reduced. In addition to the acceptable tracking ratio, root mean square error (RMSE) is used to measure the overall localization error magnitude, A lower RMSE indicates that the estimated trajectory is closer to the reference path on average.

TABLE I
ACCEPTABLE ERROR RANGE PERCENTAGES FOR RECTANGULAR PATHS.

Acceptable Error Range	Raw LSE	KalmanNet	Kalman Filter
± 200 mm	55.1%	**79.4%**	56.5%
± 500 mm	89.1%	**100.0%**	88.7%
± 1000 mm	98.2%	**100.0%**	98.3%

TABLE II
OVERALL ROOT MEAN SQUARE ERROR (RMSE) COMPARISON.

Raw LSE	KalmanNet	Kalman Filter
333.2 mm	**146.5 mm**	329 mm

Figure 6 shows the localization error distribution obtained from the raw Least Squares Estimation baseline. Compared with this baseline, Figure 7 shows that the neural network-aided Kalman filtering result keeps most estimated positions close to the reference path. This is reflected in Table I, where 79.4% of samples remain within ±200 mm, 100% remain within

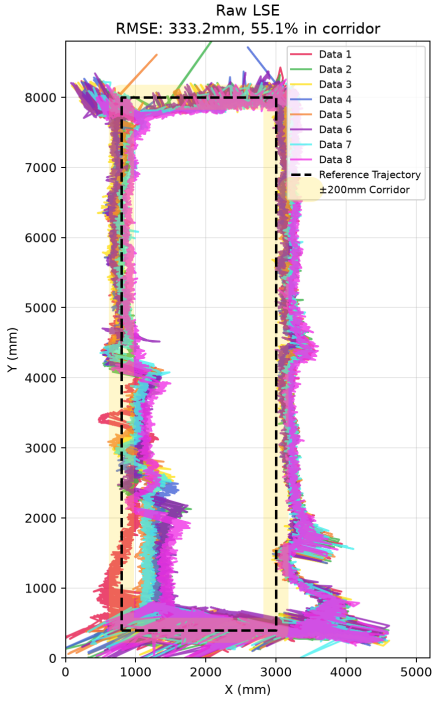


Fig. 6. Localization error evaluation for the raw Least Squares Estimation method.

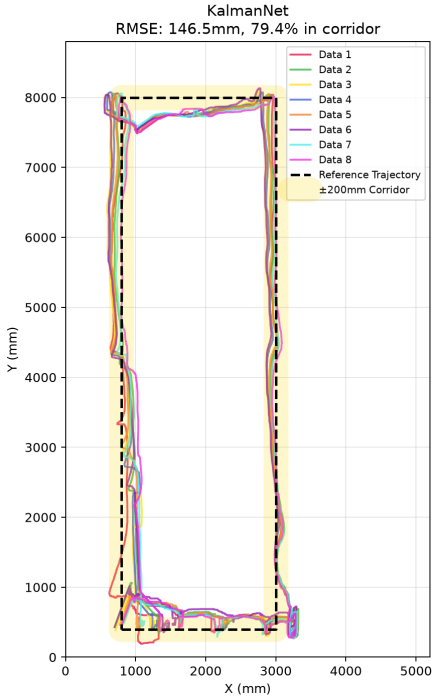


Fig. 7. Localization error evaluation for the neural network-aided Kalman filtering method.

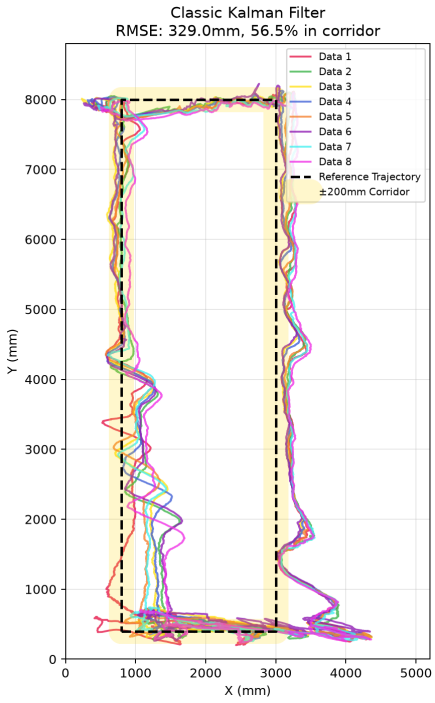


Fig. 8. Localization error evaluation for the conventional Kalman filtering method.

± 500 mm, and 100.0% remain within ± 1000 mm. The same trend appears in Table II, where KalmanNet obtains the lowest RMSE of 146.5mm. These results indicate that the estimated trajectory has fewer large deviations and follows the planned rectangular path more closely.

Figure 8 shows that the conventional Kalman filter provides less consistent tracking accuracy. As shown in Table I, 56.5% of its samples are within ± 200 mm, which is close to the 55.1% result of raw LSE. Table II also shows that the Kalman filter RMSE remains close to raw LSE, with 329 mm compared with 333.2 mm. This indicates that the conventional Kalman filter smooths part of the UWB noise but still leaves more samples farther from the reference trajectory.

The tracking results show that the system can follow the predetermined rectangular trajectory. Multipath effects and noisy UWB measurements appear as scattered or oscillatory position estimates in the baseline trajectory. The neural network-aided Kalman filtering stage is expected to reduce these effects by learning a data-driven correction behavior while preserving the recursive structure of Kalman filtering. Reporting all methods under the same trajectory, timestamps, calibration settings, and anchor layout provides a fair assessment of positioning accuracy and real-time tracking robustness.

V. Conclusion and Future Work

In this study, we presented a UWB-based indoor positioning framework for an autonomous vehicle equipped with a DWM1001C module and an onboard Raspberry Pi 4. The proposed system combines UWB ranging, least-squares position estimation, and a stabilization stage based on a neural network-aided Kalman filtering approach inspired by KalmanNet. By preserving the recursive structure of Kalman filtering while learning a data-driven gain from vehicle trajectories, the proposed method aims to improve localization accuracy, trajectory smoothness, and robustness under practical indoor conditions.

Despite these achievements, several limitations must be acknowledged. First, the experiments were conducted in a controlled laboratory environment with limited dimensions, which may not fully capture larger or more diverse real-world scenarios. Second, the system assumes anchor positions are predetermined, which may not always be feasible in dynamic or ad-hoc deployments. Future work can incorporate additional devices, including mobile devices with UWB sensors, improve energy efficiency and compatibility by incorporating BLE as an additional data source, and evaluate system performance in NLOS situations.

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